The Schwarz-Milnor lemma for braids and area-preserving diffeomorphisms

Michael Brandenbursky, Michał Marcinkowski, and Egor Shelukhin

Abstract

We prove a number of new results on the large-scale geometry of the L^p-metrics on the group of area-preserving diffeomorphisms of each orientable surface. Our proofs use in a key way the Fulton-MacPherson type compactification of the configuration space of n points on the surface due to Axelrod-Singer and Kontsevich. This allows us to apply the Schwarz-Milnor lemma to configuration spaces, a natural approach which we carry out successfully for the first time. As sample results, we prove that all right-angled Artin groups admit quasi-isometric embeddings into the group of area-preserving diffeomorphisms endowed with the L^p-metric, and that all Gambaudo-Ghys quasi-morphisms on this metric group coming from the braid group on n strands are Lipschitz. This was conjectured to hold, yet proven only for low values of n and the genus g of the surface.

Contents

1 Introduction and main results **1**
1.1 Introduction . . . 1
1.2 Preliminaries . . . 4
1.2.1 The L^p-metric . . . 4
1.2.2 Quasimorphisms . . . 5
1.2.3 Configuration spaces and braid groups . . . 5
1.2.4 Short paths and the Gambaudo-Ghys construction . . . 6
1.3 Main results . . . 7
1.4 Outline of the proof . . . 10

2 Proofs. **10**
2.1 Compactification of the configuration space . . . 10
2.2 Two metrics on $X_n(M)$ and proof of Proposition 1.1 . . . 11
2.3 Proof of Proposition 1.2 . . . 12

1 Introduction and main results

1.1 Introduction

The L^2-length of a path of volume-preserving diffeomorphisms, which describes a time-dependent flow of an ideal incompressible fluid, corresponds to the hydrodynamic action of the flow in the same way as the length of a path in a Riemannian manifold corresponds to its energy (cf. [50]). Indeed, it is the length of this path with respect to the formal right-invariant Riemannian metric on the group $\mathcal{G}$ of volume preserving diffeomorphisms introduced by Arnol'd in [1]. The L^1-length of the same path has a dynamical interpretation as the average length of a trajectory of a point under the flow.

Therefore, following the principle of least action, it makes sense to consider the infimum of the lengths of paths connecting two fixed volume-preserving diffeomorphisms. This gives rise to a right-invariant

distance function (metric) on $\mathcal{G}$. Taking the identity transformation as the initial point, Arnol'd observes that a path whose L^2-length is minimal (and equal to the distance) necessarily solves the Euler equation of an ideal incompressible fluid.

It follows from works of Ebin and Marsden [21] that for diffeomorphisms in $\mathcal{G}$ that are C^2-close to the identity, the infimum is indeed achieved. Further, more global results on the corresponding Riemannian exponential map were obtained in [19],[51] (see also [20]). In [49, 50] Shnirel'man showed, among a number of surprising facts about this subject, that in the case of the ball of dimension 3, the diameter of the L^2-metric is bounded. This result is conjectured to hold for all compact simply connected manifolds of dimension 3 or larger (see [23, 38, 2]), while its analogue in the non-simply-connected case is false [23, 7]. Furthermore, Shnirel'man has conjectured that for compact manifolds of dimension 2, the L^2-diameter is infinite.

Shnirel'man's conjecture, and its analogues for L^p-metrics, with $p \geq 1$ are by now proven. It follows from results of Eliashberg and Ratiu [23] that on compact surfaces (possibly with boundary) other than T^2 and S^2, Shnirel'man's conjecture holds for all $p \geq 1$. Their arguments rely on the Calabi homomorphism Cal [16] from the compactly supported Hamiltonian group $\mathrm{Ham}_c(M, \sigma)$ to the real numbers in the case of a surface M with non-empty boundary (σ is the area form), and on non-trivial first cohomology combined with trivial center of the fundamental group in the closed case. For the two-torus T^2 Shnirel'man's conjecture holds by [14, Appendix A]. Finally, the case of S^2 was settled in [14] by means of differential forms on the configuration space related to the cross-ratio map. In [43] the second author gave a new uniform proof of Shnirel'man's conjecture for all compact surfaces.

The methods that were used to prove Shnirel'man's conjecture are two-dimensional in nature, and have to do with braiding of trajectories of time-dependent two-dimensional Hamiltonian flows (in extended phase space). Indeed, Shnirel'man has proposed to use relative rotation numbers to bound from below the L^2-lengths of two-dimensional Hamiltonian paths in [50]. This direction is related to the method of [23] by a theorem of Fathi [24] and Gambaudo and Ghys [27] (see also [48, 31, 33]). This theorem shows that the Calabi homomorphism is proportional to the relative rotation number of the trajectories of two distinct points in the two-disc $\mathbb{D}$ under a Hamiltonian flow, averaged over the configuration space of ordered pairs of distinct points (x_1, x_2) in the two-disc.

This line of research was notably pursued in [29], and further in [4], [18], [7], [9], [39], [14] obtaining quasi-isometric and bi-Lipschitz embeddings of various groups (right-angled Artin groups and additive groups of finite-dimensional real vector spaces) into $\mathrm{Ham}_c(\mathbb{D}^2, dx \wedge dy)$, into $\ker(\mathrm{Cal}) \subset \mathrm{Ham}_c(\mathbb{D}^2, dx \wedge dy)$, and into $\mathrm{Ham}(S^2, \sigma)$ endowed with their respective L^p-metrics (see [10] for similar embedding results on manifolds with a sufficiently complicated fundamental group). In all cases, the key technical estimate is an upper bound, via the L^p-length of an isotopy of volume-preserving diffeomorphisms, of the average, over all points in a configuration space of the manifold, of the word length in the fundamental group of the trace of the point under the induced isotopy (closed up to a loop by a system of short paths on the configuration space).

Such estimates were initially produced by means of analyzing relative rotation numbers of pairs of braids, or quadruples as in [14]. However, in the case of a single braid, as observed by Polterovich, a simpler estimate is possible via the Schwarz-Milnor lemma[1] [22, 54, 44]. Indeed, according to this result, the universal cover of compact proper length space, in this case our surface, is quasi-isometric to its fundamental group. In this way one can control the word length of trajectories in terms of their lengths, which allows one to prove the required upper bound. A similar estimate in the case of braids on more than one strand is not as readily available, because the configuration space $X_n(M)$ of n points on M is not a compact metric space. The arguments of the second author in [43] rely crucially on a new complete metric on $X_n(M)$. However, with this metric $X_n(M)$ still can not be considered to be a compact metric space from the point of view of the Schwarz-Milnor lemma.

[1] "Schwarz" is a transliteration from Russian; alternative spellings "Švarc", "Schwartz" sometimes appear in the literature.

In this paper we show how to successfully carry out the strategy of the Schwarz-Milnor lemma for configuration spaces. While hints of a similar approach can be discerned in [26] in the special case of the two-disc and of double collisions, it was not known earlier to be applicable in the general context discussed herein. Specifically, we consider a metric on $X_n(M)$ coming from a natural compactification $\overline{X}_n(M)$ thereof, which is a compact geodesic metric space. This compactification is equivariant under the action of the diffeomorphism group $\mathrm{Diff}(M)$ on $X_n(M)$ in the sense that the action extends naturally to the compactification[2]. Furthermore, the map $\pi_1(X_n(M)) \to \pi_1(\overline{X}_n(M))$ induced by the inclusion is an isomorphism. Hence we may apply the Schwarz-Milnor lemma to $\overline{X}_n(M)$. This, and further comparison to the metric from [43] allows us to prove our main estimate.

The compactification we use was introduced by Axelrod-Singer in [3] and by Kontsevich in [40, 41], inspired by the Fulton-MacPherson compactification in algebraic geometry [25]. The compactification $\overline{X}_n(M)$ is roughly speaking a certain positive oriented blow-up of M^n along its multi-diagonals. For instance, one part of its codimension 1 boundary stratum is identified with the disjoint union of spaces of the form $N_1(D_{ij}(M))|_{D^0_{ij}(M)}$ where N_1 is the unit normal bundle, $D_{ij}(M) \subset M^n$ is the submanifold of points $(x_1, \ldots, x_n)$ where $x_i = x_j$ and $D^0_{ij}(M) \subset D_{ij}(M)$ is the open dense subset where $x_k \neq x_i = x_j$ for all $k \in \{1, \ldots, n\} \setminus \{i, j\}$. Note that this is an S^1-bundle. Intuitively, from a physical perspective, this means that one resolves a double collision of points by recording the collision point and the direction in which they have collided. Other parts of the codimension 1 stratum correspond to simple k-tuple collisions, and correspond to S^{2k-3}-bundles normal to k-diagonals. Higher strata correspond to more complicated collisions modeled by suitable graphs. It will, however, be technically most convenient for us to use a model of this compactification recently constructed directly as a subspace of a Euclidean space by Sinha [52]. We describe this construction in Section 2 below.

Finally, we observe that lower bounds on the average word length can often be provided by quasimorphisms - functions that are additive with respect to the group multiplication - up to an error which is uniformly bounded (as a function of two variables). The quasimorphisms we consider here were introduced and studied by Gambaudo and Ghys in the beautiful paper [28] (see also [8, 45, 46, 47, 6, 14, 13]). These quasimorphisms essentially appear from invariants of braids traced out by the action of a Hamiltonian path on an ordered n-tuple of distinct points in the surface (suitably closed up), averaged over the configuration space $X_n(M)$ of n-tuples of distinct points on the surface M. As one of our results, we prove that all homogeneous Gambaudo-Ghys quasi-morphisms are Lipschitz in the L^p-metric for all $p \geq 1$. This subsumes all previous results in this direction and provides a maximally general result. It also contrasts a recent result of Khanevsky according to which none of these quasi-morphisms are continuous in Hofer's metric [37].

We prove further stronger results on the large-scale geometry on the L^p-metric on $\mathcal{G} = \mathrm{Diff}_{c,0}(M, \sigma)$ for a compact surface (M, σ) with an area form. In particular, we provide bi-Lipschitz group monomorphisms of $\mathbb{R}^m$ endowed with the standard (say Euclidean) metric into $(\mathcal{G}, d_{L^p})$ for each positive integer m and each $p \geq 1$. Finally, our methods combined with an argument of Kim-Koberda [39] (cf. Crisp-Wiest [18], Benaim-Gambaudo [4]) show the existence of quasi-isometric group monomorphisms from each right-angled Artin group to $(\mathcal{G}, d_{L^p})$ for each $p \geq 1$. We note that this was previously known only for $\mathbb{D}^2$ and S^2 [39, 14].

Let $M_1 \hookrightarrow M_2$ be a measure preserving embedding of surfaces. It is an open question if the induced Lipschitz monomorphism $\mathrm{Diff}_0(M_1, \sigma_1) \hookrightarrow \mathrm{Diff}_0(M_2, \sigma_2)$ is a quasi-isometric embedding. Note that our results provide a partial positive answer to this problem, see Remark 1.3.5.

Acknowledgments.

MB was partially supported by Humboldt research fellowship. MM was partially supported by NCN (Sonatina 2018/28/C/ST1/00542) and by the GAČR project 19-05271Y and by RVO: 67985840.

[2]It is curious to note that the same is *not* true for the action of $\mathrm{Homeo}(M)$ as was recently proven [42].

ES was partially supported by an NSERC Discovery Grant, by the Fondation Courtois, and by a Sloan Research Fellowship.

1.2 Preliminaries

1.2.1 The L^p-metric

Let M denote a smooth oriented manifold without boundary that is either closed, or $M = X \setminus \partial X$ for a compact manifold X. Let M be endowed with a Riemannian metric g and smooth measure μ (given by an orientation on M and volume form, which in our case of a surface M is an area form σ). We require that g and μ extend continuously to X in the second case. Finally denote by

$$\mathcal{G} = \mathrm{Diff}_{c,0}(M, \mu)$$

the identity component of the group of compactly supported diffeomorphisms of M preserving the smooth measure μ. In other words, if $M = X \setminus \partial X$, it is the identity component of the group of measure preserving diffeomorphisms of X fixing point-wise a neighbourhood of ∂X.

Fix $p \geq 1$. For a smooth isotopy $\{\phi_t\}_{t\in[0,1]}$, from $\phi_0 = 1$ to $\phi_1 = \phi$, we define the L^p-length by

$$l_p(\{\phi_t\}) = \int_0^1 \left(\frac{1}{\mathrm{vol}(M,\mu)} \cdot \int_M |X_t|^p d\mu \right)^{\frac{1}{p}} dt\,,$$

where $X_t = \frac{d}{dt'}|_{t'=t}\phi_{t'} \circ \phi_t^{-1}$ is the time-dependent vector field generating the isotopy $\{\phi_t\}$, and $|X_t|$ is its length with respect to the Riemannian structure on M. As is easily seen by a displacement argument, the L^p-length functional determines a non-degenerate norm on $\mathcal{G}$ by the formula

$$d_p(\mathbf{1}, \phi) = \inf\ l_p(\{\phi_t\}).$$

This in turn defines a right-invariant metric on $\mathcal{G}$ by the formula

$$d_p(\phi_0, \phi_1) = d_p(\mathbf{1}, \phi_1 {\phi_0}^{-1}).$$

Remark 1.2.1. Consider the case $p = 1$. It is easy to see that the L^1-length of an isotopy is equal to the average Riemannian length of the trajectory $\{\phi_t(x)\}_{t\in[0,1]}$ (over $x \in M$, with respect to μ). Moreover for each $p \geq 1$, by Jensen's (or Hölder's) inequality, we have

$$l_p(\{\phi_t\}) \geq l_1(\{\phi_t\}).$$

Denote by $\widetilde{\mathbf{1}}$ the identity element of the universal cover $\widetilde{\mathcal{G}}$ of $\mathcal{G}$. Similarly one has the L^p-pseudo-norm (that induces the right-invariant L^p-pseudo-metric) on $\widetilde{\mathcal{G}}$, defined for $\widetilde{\phi} \in \widetilde{\mathcal{G}}$ as

$$d_p(\widetilde{\mathbf{1}}, \widetilde{\phi}) = \inf\ l_p(\{\phi_t\}),$$

where the infimum is taken over all paths $\{\phi_t\}$ in the class of $\widetilde{\phi}$. Clearly $d_p(\mathbf{1}, \phi) = \inf d_p(\widetilde{\mathbf{1}}, \widetilde{\phi})$, where the infimum runs over all $\widetilde{\phi} \in \widetilde{\mathcal{G}}$ that map to ϕ under the natural epimorphism $\widetilde{\mathcal{G}} \to \mathcal{G}$.

Up to bi-Lipschitz equivalence of metrics (d and d' are equivalent if $\frac{1}{C}d \leq d' \leq Cd$ for a certain constant $C > 0$) the L^p-metric on $\mathcal{G}$ (and its pseudo-metric analogue on $\widetilde{\mathcal{G}}$) is independent of the choice Riemannian structure and of the volume form μ on M. In particular, the question of boundedness or unboundedness of the L^p-metric enjoys the same invariance property.

Terminology: For a positive integer n, we use $A, B, C > 0$ as generic notation for positive constants that depend only on M, μ, g and n.

1.2.2 Quasimorphisms

Some of our results have to do with the notion of a quasimorphism. Quasimorphisms are a helpful tool for the study of non-abelian groups, especially those that admit few homomorphisms to $\mathbb{R}$. A quasimorphism $r : G \to \mathbb{R}$ on a group G is a real-valued function that satisfies

$$r(xy) = r(x) + r(y) + b_r(x, y),$$

for a function $b_r : G \times G \to \mathbb{R}$ that is uniformly bounded:

$$\delta(r) := \sup_{G \times G} |b_r| < \infty.$$

A quasimorphism $\overline{r} : G \to \mathbb{R}$ is called *homogeneous* if $\overline{r}(x^k) = k\overline{r}(x)$ for all $x \in G$ and $k \in \mathbb{Z}$. In this case, it is additive on each pair $x, y \in G$ of commuting elements: $r(xy) = r(x) + r(y)$ if $xy = yx$.

For each quasimorphism $r : G \to \mathbb{R}$ there exists a unique homogeneous quasimorphism $\overline{r}$ that differs from r by a bounded function:

$$\sup_G |\overline{r} - r| < \infty.$$

It is called the *homogenization* of r and satisfies

$$\overline{r}(x) = \lim_{n \to \infty} \frac{r(x^n)}{n}.$$

Denote by $Q(G)$ the real vector space of homogeneous quasimorphisms on G.

For a finitely-generated group G, with finite symmetric generating set S, define the word norm $|\cdot|_S : G \to \mathbb{Z}_{\geq 0}$ by

$$|g|_S = \min\{k \,|\, g = s_1 \cdot \ldots \cdot s_k, \, \forall\, 1 \leq j \leq k, \, s_j \in S\}$$

for $g \in G$. This is a norm on G, and as such it induces a right-invariant metric $d_S : G \times G \to \mathbb{Z}_{\geq 0}$ by $d_S(f, g) = |gf^{-1}|_S$. This metric is called the word metric. In this setting, any quasimorphism $r : G \to \mathbb{R}$ is controlled by the word norm. Indeed, for all $g \in G$,

$$|r(g)| \leq \left(\delta(r) + \max_{s \in S} |r(s)|\right) \cdot |g|_S.$$

When a specific symmetric generating set S for G can be fixed, we will usually denote $|\cdot|_S$ by $|\cdot|_G$.

We refer to [17] for more information about quasimorphisms.

1.2.3 Configuration spaces and braid groups

For a manifold M, which in this paper is usually of dimension 2, the configuration space $X_n(M) \subset M^n$ of n-tuples of points on M is defined as

$$X_n(M) = \{(x_1, \ldots, x_n) |\, x_i \neq x_j, \, 1 \leq i < j \leq n\}.$$

That is

$$X_n(M) = M^n \setminus \bigcup_{1 \leq i < j \leq n} D_{ij}$$

where for $1 \leq i < j \leq n$, the partial diagonal $D_{ij} \subset M^n$ is defined as $D_{ij} = \{(x_1, \ldots, x_n) |\, x_i = x_j\}$. Note that D_{ij} is a submanifold of M^n of codimension $\dim M$.

Finally, if $\dim M = 2$, we define the pure braid group of M as

$$P_n(M) = \pi_1(X_n(M)).$$

Noting that the symmetric group S_n on n elements acts freely on $X_n(M)$, we form the quotient $C_n(M) = X_n(M)/S_n$ and define the full braid group of M as

$$B_n(M) = \pi_1(C_n(M)).$$

We note that $P_n(M)$ and $B_n(M)$ enter the exact sequence $1 \to P_n(M) \to B_n(M) \to S_n \to 1$. In particular $P_n(M)$ is a normal subgroup of $B_n(M)$ of finite index. We refer to [36] for further information about braid groups.

1.2.4 Short paths and the Gambaudo-Ghys construction

Let M be a compact oriented surface. Given a real valued quasimorphism r on $P_n(M) = \pi_1(X_n(M), q)$ for a fixed basepoint $q \in X_n(M)$ there is a natural way to construct a real valued quasimorphism on the universal cover $\widetilde{\mathcal{G}}$ of the group $\mathcal{G} = \mathrm{Diff}_{c,0}(M, \sigma)$ of area preserving diffeomorphisms of the surface M. We shall see that in the case of $M \neq T^2$ this induces a quasimorphism on $\mathcal{G}$ itself, because the fundamental group of $\mathcal{G}$ is finite. The same is true for $M = T^2$ where we consider the group $\mathcal{G} = \mathrm{Ham}(M, \sigma)$ of Hamiltonian diffeomorphisms instead. This is not a restrictive condition from the viewpoint of large-scale geometry, since by a small modification of [14, Proposition A.1], the inclusion $(\mathrm{Ham}(T^2, \sigma), d_{L^p}) \hookrightarrow (\mathrm{Diff}_0(T^2, \sigma), d_{L^p})$ is a quasi-isometry for all $p \geq 1$.

The construction is carried out by the following steps (cf. [28, 45, 6]).

1. For all $x \in X_n(M) \setminus Z$, with Z a closed negligible subset (e.g. a union of submanifolds of positive codimension) choose a smooth path $\gamma(x) : [0,1] \to X_n(M)$ between the basepoint $q \in X_n(M)$ and x. Make this choice continuous in $X_n(M) \setminus Z$. We first choose a system of paths on M itself. Then we consider the induced coordinate-wise paths in M^n, and pick Z to ensure that these induced paths actually lie in $X_n(M)$. After choosing the system of paths $\{\gamma(x)\}_{x \in X_n(M) \setminus Z}$ we extend it measurably to all $x \in X_n(M)$ (obviously, no numerical values computed in the paper will depend on this extension). We call the resulting choice a "system of short paths".

2. Given a path $\{\phi_t\}_{t \in [0,1]}$ in $\mathcal{G}$ starting at Id, and a point $x \in X_n(M)$ consider the path $\{\phi_t \cdot x\}$, to which we then catenate the corresponding short paths. That is consider the loop

   $$\lambda(x, \{\phi_t\}) := \gamma(x) \# \{\phi_t \cdot x\} \# \gamma(y)^{-1}$$

   in $X_n(M)$ based at q, where $^{-1}$ denotes time reversal. Hence we obtain for each $x \in X_n(M) \setminus Z \cup (\phi_1)^{-1}(Z)$ an element $[\lambda(x, \{\phi_t\})] \in \pi_1(X_n(M), q)$ (or rather for each $x \in X_n(M)$ after the measurable extension in Step 1).

3. Consequently applying the quasimorphism $r : \pi_1(X_n(M), q) \to \mathbb{R}$ we obtain a measurable function $f : X_n(M) \to \mathbb{R}$. Namely $f(x) = r([\lambda(x, \{\phi_t\})])$. The quasimorphism Φ on $\widetilde{\mathcal{G}}$ is defined by

   $$\Phi([\{\phi_t\}]) = \int_{X_n(M)} f \, d\mu^{\otimes n}.$$

   It is immediate to see that this function is well-defined by topological reasons. The quasimorphism property follows by the quasimorphism property of r combined with finiteness of volume. The fact that the function f is absolutely integrable can be shown to hold a-priori by a reduction to the case of the disc. We note, however, that by Tonelli's theorem this fact follows as a by-product of the proof of our main theorem, and therefore requires no additional proof.

4. Of course this quasimorphism can be homogenized, to obtain a homogeneous quasimorphism $\overline{\Phi}$.

Remark 1.2.2. If $M \neq S^2$ and $M \neq T^2$, then $\pi_1(\mathcal{G}) = 0$ and for $M = T^2$ the same is true for $\mathcal{G} = \mathrm{Ham}(M, \sigma)$. In the case $M = S^2$, by the result of Smale [53] $\pi_1(\mathcal{G}) = \mathbb{Z}/2\mathbb{Z}$. Hence the quasimorphisms descend to quasimorphisms on $\mathcal{G}$, e.g. by minimizing over the two-element fibers of the projection $\widetilde{\mathcal{G}} \to \mathcal{G}$. For $\overline{\Phi}$, the situation is easier since by homogeneity it vanishes on $\pi_1(\mathcal{G}) \subset Z(\widetilde{\mathcal{G}})$, and therefore

depends only on the image in $\mathcal{G}$ of an element in $\widetilde{\mathcal{G}}$. We keep the same notations for the induced quasimorphisms.

1.3 Main results

Recall that we work with the group $\mathcal{G} = \mathrm{Diff}_{c,0}(M, \sigma)$ for (M, σ) a compact oriented surface with an area form. Our main technical result is the following.

Theorem 1. *For an isotopy $\overline{\phi} = \{\phi_t\}$ in $\mathcal{G}$, the average word norm of a trajectory $\lambda(x, \overline{\phi})$ is controlled by the L^1-length of $\overline{\phi}$:*

$$W(\overline{\phi}) = \int_{X_n(M)\setminus Z \cup (\phi_1)^{-1}(Z)} |[\lambda(x, \overline{\phi})]|_{P_n(M)} \, d\mu^{\otimes n}(x) \leq A \cdot l_1(\overline{\phi}) + B,$$

for certain constants $A, B > 0$.

Remark 1.3.1. Note that $W(\overline{\phi})$ depends only on the class $\widetilde{\phi} = [\overline{\phi}] \in \widetilde{\mathcal{G}}$ of $\overline{\phi}$ in the universal cover $\widetilde{\mathcal{G}}$ of $\mathcal{G}$. Hence for all compact oriented surfaces other than S^2 and T^2, $W(\overline{\phi})$ does not depend on the chosen isotopy $\overline{\phi}$, but only on the diffeomorphism ϕ_1. This is because $\mathcal{G}$ is simply connected.

Theorem 1 has a number of consequences concerning the large-scale geometry of the L^1-metric on $\mathcal{G}$. Firstly, as any quasimorphism on a finitely generated group is controlled by the word norm, we immediately obtain the following statement.

Corollary 1. *The homogenization $\overline{\Phi}$ of each Gambaudo-Ghys quasimorphism Φ satisfies*

$$|\overline{\Phi}(\phi)| \leq C \cdot d_1(\phi, 1).$$

In particular, Corollary 1 implies that all Gambaudo-Ghys quasimorphisms are continuous in the L^1-metric (and hence in the L^p-metric, see Remark 1.3.4), a fact which was known so far only for the genus zero case (see [14, 7]) and for the higher genus case only when one considers Gambaudo-Ghys quasimorphisms coming from the fundamental group $P_1(M)$, see [7]. Note that none of these quasimorphisms are continuous in the Hofer metric by a recent result by Khanevsky [37].

By a theorem of Ishida [34] in the genus zero case and its generalisation to any compact oriented surface [12, Theorem 2.2] (see all well Brandenbursky-Kedra-Shelukhin [11] in the genus one case, and Brandenbursky [8] in the higher genus case), the image of the map $Q(P_n(M)) \xrightarrow{GG} Q(\mathcal{G})$ is infinite dimensional for $n \geq 4$. Thus $Q(\mathcal{G})$ is an infinite-dimensional vector space. Hence by Corollary 1 we obtain in particular the following.

Corollary 2. *The L^1-diameter of $\mathcal{G}$ is infinite.*

Let Ent^k be the set of products of at most k entropy zero diffeomorphisms. Theorem 1 in [12] and Corollary 1 imply the following.

Corollary 3. *For each positive integer k, the complement in $\mathcal{G}$ of the set Ent^k contains a ball of any arbitrarily large radius in the L^1-metric. In particular, the set of non-autonomous diffeomorphisms contains a ball of any arbitrarily large radius in the L^1-metric.*

In what follows, we apply an argument of Kim-Koberda [39] (cf. Benaim-Gambaudo [4] and Crisp-Wiest [18]), and use Theorem 1 in order to obtain the following statement, which generalizes an answer to a question of Kapovich [35] in the case of S^2 [14] to arbitrary compact oriented surfaces.

Corollary 4. *The metric group $(\mathcal{G}, d_1)$ admits a quasi-isometric group embedding from each right-angled Artin group endowed with the word metric.*

Proof. Assume first that $M = \Sigma_g$ a closed surface of a positive genus. The case of S^2 was already done in [14] and the case of surfaces with boundary will be discussed in the end of the proof. Let $n \in \mathbb{N}$. Let $D \subset \Sigma_g$ be a smoothly embedded open disc and $z_1, \dots, z_n$ disjoint points in D. For each i, let $D_i \subset D$ be an embedded open disc centered at z_i such that each short geodesic between two points in D_i lies in D_i, and $D_i \cap D_j = \emptyset$ for each $i \neq j$. We denote by $\mathrm{Diff}(D; D_1, \dots, D_n) < \mathcal{G}$ the group which consists of all diffeomorphisms in $\mathcal{G}$ which are compactly supported in D and act by identity on each D_i. This subgroup is equipped with the L^1-metric coming from $\mathcal{G}$.

Lemma 1. *Let $n > 4$. We identify P_n with the pure mapping class group of the disc D punctured at $z_1, \dots, z_n$. Then the inclusion $P_n \to P_n(\Sigma_g)$ is a quasi-isometric embedding.*

Proof. Since P_n is quasi-isometric to B_n and $P_n(\Sigma_g)$ is quasi-isometric to $B_n(\Sigma_g)$, it is enough to show that the inclusion $B_n \to B_n(\Sigma_g)$ is a quasi-isometric embedding, where B_n is identified with the mapping class group of the disc D punctured at $z_1, \dots, z_n$. By results of Goldberg [30] (c.f. Birman [5]) we have that B_n is a subgroup of $B_n(\Sigma_g)$. In addition, the composition

$$B_n \xrightarrow{i} B_n(\Sigma_g) \xrightarrow{F} MCG_{g,n}$$

is injective, where i is the inclusion, the map $F : B_n(\Sigma_g) \to MCG_{g,n}$ is the point pushing map and $MCG_{g,n}$ is the mapping class group of Σ_g punctured at $z_1, \dots, z_n$. Moreover, by a result of Hamenstadt [32, Theorem 2] the map $F \circ i : B_n \hookrightarrow MCG_{g,n}$ is a quasi-isometric embedding for $n > 4$. Since F is Lipschitz, it follows follows that i is a quasi-isometric embedding (see [10, Lemma 2.1]). $\square$

Let us return to the proof of the corollary. Consider the homomorphism

$$H : \mathrm{Diff}(D; D_1, \dots, D_n) \to P_n < P_n(\Sigma_g),$$

where $H(\phi)$ is an element in $P_n(\Sigma_g)$ represented by the path $(\phi_t(z_1), \dots, \phi_t(z_n))$ in the configuration space, and $\{\phi_t\}$ is any isotopy in $\mathcal{G}$ between the identity ϕ_0 and $\phi := \phi_1$.

It follows from the results of Kim-Koberda [39] (c.f. Crisp-Wiest [18] and Benaim-Gambaudo [4]) that for each RAAG Γ there exists n, which can be assumed to satisfy $n > 4$, and an embedding of Γ into $\mathrm{Diff}(D; D_1, \dots, D_n)$ whose composition with H is a quasi-isometric embedding of Γ into P_n. By Lemma 1, it gives as well a quasi-isometric embedding of Γ into $P_n(\Sigma_g)$. Then in order to obtain a quasi-isometric embedding of Γ into $\mathcal{G}$ it is enough to show that the map H is large-scale Lipschitz whenever $n > 4$.

This fact follows from our main result, Theorem 1. Let us prove it. Set

$$X(D_1, \dots, D_n) := \{(x_1, \dots, x_n) \in X_n(\Sigma_g) |\, x_i \in D_i\}.$$

Let $\{\phi_t\}$ be any isotopy in $\mathcal{G}$ between the identity $\phi_0 = \mathbf{1}$ and $\phi := \phi_1$. It follows from Theorem 1 that there exist positive constants A and B which depend only on n such that

$$\int_{X(D_1,\dots,D_n)} |[\lambda(x,\phi)]|_{P_n(\Sigma_g)}\, d\mu^{\otimes n}(x) \leq A \cdot l_1(\phi) + B\,.$$

Note that for each $x \in X(D_1, \dots, D_n)$ we have $[\lambda(x, \phi)] = H(\phi)$. Thus

$$\int_{X(D_1,\dots,D_n)} |[\lambda(x,\phi)]|_{P_n(\Sigma_g)}\, d\mu^{\otimes n}(x) = \mathrm{vol}(X(D_1, \dots, D_n)) \cdot |H(\phi)|_{P_n(\Sigma_g)},$$

and we obtain the result, i.e., H is large-scale Lipschitz:

$$|H(\phi)|_{P_n(\Sigma_g)} \leq \frac{A}{\mathrm{vol}(X(D_1, \dots, D_n))} \cdot l_1(\phi) + \frac{B}{\mathrm{vol}(X(D_1, \dots, D_n))}\,.$$

Assume now that M_1 is a surface with boundary. We can embed M_1 in a closed surface $M_2 = \Sigma_g$ such that the embedding is area preserving. This induces a monomorphism $\iota\colon \mathrm{Diff}_{c,0}(M_1, \sigma_1) \hookrightarrow \mathrm{Diff}_{c,0}(M_2, \sigma_2)$ (note that elements of $\mathrm{Diff}_{c,0}(M_1, \sigma_1)$ fix point-wise the neighbourhood of the boundary, so they can be extended by the identity). Since ι it Lipschitz, and the embedding of Γ to $\mathrm{Diff}_{c,0}(M_2, \sigma_2)$ can be arranged to factor through $\mathrm{Diff}_{c,0}(M_1, \sigma_1)$, we get a quasi-isometric embedding to $\mathrm{Diff}_{c,0}(M_1, \sigma_1)$ (see [10, Lemma 2.1]). □

Remark 1.3.2. We note that Corollary 4 implies Corollary 2, providing the latter with a proof that does not use quasimorphisms.

Furthermore, note that Corollary 4 implies that for each $k \in \mathbb{N}$ there is a quasi-isometric embedding $i_k : \mathbb{Z}^k \to (\mathcal{G}, d_1)$. Moreover, there exists a k-tuple of autonomous Hamiltonian flows (one-parameter subgroups) $\{\{\phi_i^t\}_{t\in\mathbb{R}}\}_{1\le i\le k}$ which have disjoint supports, and $i_k(e_j) = \phi_j^1$ for each $1 \le j \le k$, where $e_j = (0, \dots, 1, \dots, 0)$ and 1 lies in the j-th entry. The above k-tuple of flows defines a homomorphism $j_k : \mathbb{R}^k \to (\mathcal{G}, d_1)$ such that $i_k = j_k|_{\mathbb{Z}^k}$. Since $\mathbb{R}^k$ is quasi-isometric to $\mathbb{Z}^k$, we obtain the following statement:

Corollary 5. *The metric group $(\mathcal{G}, d_1)$ admits a quasi-isometric embedding from $(\mathbb{R}^k, d)$ where d is any metric on $\mathbb{R}^k$ induced by a vector-space norm.*

The following lemma allows us to prove that the quasi-isometric embeddings of $\mathbb{Z}^k, \mathbb{R}^k$ from Corollary 5 are in fact bi-Lipschitz embeddings.

Lemma 2. *Let (G, d) be a metric group and let $A \subset V$ be a subgroup of a normed linear space. Then each Lipschitz quasi-isometric embedding $j : A \to G$ of metric groups is a bi-Lipschitz embedding. If A is discrete and finitely generated then each homomorphism $j : A \to G$ is Lipschitz.*

Proof. Let $j : A \to G$ be a homomorphism. If A is discrete and finitely generated, then $A \cong \mathbb{Z}^l$ for $l \in \mathbb{Z}_{\ge 0}$, the norm being equivalent to the standard norm on $\mathbb{Z}^l$. The upper bound $d(j(x), \mathbf{1}) \le C|x|$ for all $x \in A$ is now immediate. Hence j is Lipschitz.

Suppose now that $A \subset V$ is a subgroup and $j : A \to G$ is a Lipschitz quasi-isometric embedding. Let $C \ge 1$ be such that

$$\frac{1}{C}|x| - B \le d(j(x), \mathbf{1}) \le C|x| \tag{1}$$

for a constant $B \ge 0$ and all $x \in A$. We claim that the inequality (1) holds with $B = 0$. Indeed, consider (1) for x^m where $m \in \mathbb{Z}_{>0}$. We have

$$\frac{1}{C}m|x| - B \le d(j(x^m), \mathbf{1}) = d(j(x)^m, \mathbf{1}) \le m \cdot d(j(x), \mathbf{1}),$$

where the last inequality is due to the right-invariance of d. Dividing by m yields

$$\frac{1}{C}|x| - B/m \le d(j(x), \mathbf{1}),$$

which finishes the proof by taking limits as $m \to \infty$. □

This immediately implies the following strengthening of Corollary 5, since j_k is evidently Lipschitz.

Corollary 6. *The metric group $(\mathcal{G}, d_1)$ admits a bi-Lipschitz embedding from $(\mathbb{R}^k, d)$ where d is any metric on $\mathbb{R}^k$ induced by a vector-space norm.*

Remark 1.3.3. It should be possible to prove Corollary 6 by using quasimorphisms, as in [14].

Remark 1.3.4. Let $p \geq 1$. Note that since, by Jensen's (or Hölder's) inequality,

$$d_1 \leq d_p,$$

all the above results for d_1 continue to hold for d_p.

Recall that if M has boundary then we assume that elements of $\mathrm{Diff}_{c,0}(M, \sigma_1)$ fix point-wise an open neighbourhood of ∂M. Thus if $(M_1, \sigma_1) \hookrightarrow (M_2, \sigma_2)$ is a measure preserving embedding of manifolds, then extending a diffeomorphism on M_1 by the identity to a diffeomorphim of M_2 gives a well defined monomorphism $\iota\colon \mathrm{Diff}_{c,0}(M_1, \sigma_1) \hookrightarrow \mathrm{Diff}_{c,0}(M_2, \sigma_2)$.

We finish this section with a question.

Question 1. Let $(M_1, \sigma_1) \hookrightarrow (M_2, \sigma_2)$ be a measure preserving embedding of surfaces. Is the monomorphism $\iota\colon \mathrm{Diff}_{c,0}(M_1, \sigma_1) \hookrightarrow \mathrm{Diff}_{c,0}(M_2, \sigma_2)$ a quasi-isometric embedding?

Remark 1.3.5. Note that this question is motivated by our results since the statement holds for the compositions of ι with the embeddings of right-angled Artin groups and $(\mathbb{R}^k, d)$ from the proofs of Corollary 4 and Corollary 6.

1.4 Outline of the proof

In order to show Theorem 1, we define a Riemannian metric g on $X_n(M)$ such that d_g, the geodesic metric on $X_n(M)$ induced by g, extends to the geodesic metric on the compactification $\overline{X}_n(M)$. This allows us to use the Schwarz-Milnor lemma.

More precisely, Theorem 1 is a consequence of the following two propositions (we defer the proofs to Section 2).

Proposition 1.1. *Let λ be a piecewise C^1 loop in $X_n(M)$ based at q. Let S be a finite generating set of $P_n(M)$. The word norm of the class $[\lambda] \in \pi_1(X_n(M), q) \cong P_n(M)$ with respect to S satisfies*

$$|[\lambda]|_{P_n(M)} \leq A_0 \cdot l_g(\lambda) + B_0,$$

for constants $A_0, B_0 > 0$ depending only on S and n.

The next proposition says that the average length of loops $\lambda(x, \overline{\phi})$ is controlled by $l_1(\overline{\phi})$.

Proposition 1.2. *Let $\overline{\phi} = \{\phi_t\}$ be an isotopy in $\mathcal{G}$ such that $\phi_0 = 1$. There exist constants $A_1, B_1 > 0$ depending only on n, such that*

$$\int_{X_n(M) \setminus Z \cup (\phi_1)^{-1}(Z)} l_g(\lambda(x, \overline{\phi}))\, d\mu^{\otimes n}(x) \leq A_1 \cdot l_1(\overline{\phi}) + B_1.$$

To show this proposition, we first compare a metric g to an auxiliary metric g_0. The metric g_0 does not extend to a metric on the compactification $\overline{X}_n(M)$, but it is relatively easy to show [43, Lemma 5.2], that the inequality from Proposition 1.2 holds for g_0.

2 Proofs.

2.1 Compactification of the configuration space

In this section M is a compact m-dimensional manifold. Below we describe the compactification of $X_n(M)$ using an embedding of the configuration space into a high-dimensional Euclidean space. We follow closely the construction given in [52].

Let us start with describing a family of maps on $X_n(\mathbb{R}^d)$. Let $\pi_{ij}\colon X_n(\mathbb{R}^d) \to S^{d-1}$ by defined by the formula

$$\pi_{ij}(x) = \frac{x_i - x_j}{|x_i - x_j|},$$

where $0 \leq i < j \leq n$ and $x = (x_1, \ldots, x_n) \in X_n(\mathbb{R}^d)$. Let $s_{ijk}\colon X_n(\mathbb{R}^d) \to [0, \infty]$ be defined by the formula

$$s_{ijk}(x) = \frac{|x_i - x_j|}{|x_i - x_k|},$$

where $0 \leq i < j < k \leq n$ and $[0, \infty]$ is the one point compactification of $[0, \infty)$. Let $\iota\colon X_n(\mathbb{R}^d) \hookrightarrow (\mathbb{R}^d)^n$ be the standard inclusion and let $A_n[\mathbb{R}^d] = (\mathbb{R}^d)^n \times (S^{n-1})^{\binom{n}{2}} \times [0, \infty]^{\binom{n}{3}}$. We consider the embedding of $X_n(\mathbb{R}^d)$ to the ambient space $A_n[\mathbb{R}^d]$ given by the product map

$$\alpha_n = \iota \times (\pi_{ij}) \times (s_{ijk})\colon X_n(\mathbb{R}^d) \to A_n[\mathbb{R}^d].$$

Suppose M is a submanifold of $\mathbb{R}^d$. Let $\overline{X}_n(M)$ be the closure of the image $\alpha_n(X_n(M))$ in $A_n[\mathbb{R}^d]$. Then $\overline{X}_n(M)$ is a manifold with boundary, and the inclusion $X_n(M) \hookrightarrow \overline{X}_n(M)$ is a homotopy equivalence [52, Theorem 4.4 and Corollary 4.5].

2.2 Two metrics on $X_n(M)$ and proof of Proposition 1.1

On $[0, \infty]$ we introduce the structure of a smooth manifold by a 1-map atlas $e^{-x}\colon [0, \infty] \to [0, 1]$. Let g_{exp} be the Riemannian metric on $[0, \infty]$ given by the pull-back of the Euclidean metric from $[0, 1]$. In particular, we have that $|\partial_x|_{g_{exp}} = e^{-x}$, where ∂_x is a standard 'unit' vector field on $[0, \infty)$.

Let *euc* be the metric on $A_n[\mathbb{R}^d]$ given by the product of standard metrics on $\mathbb{R}^d$, S^{d-1} and $([0, \infty], g_{exp})$. The first metric on $X_n(\mathbb{R}^d)$ we want to consider is defined to be the pull-back of *euc* to $X_n(\mathbb{R}^d)$ by the map α_n:

$$g = \alpha_n^*(euc).$$

Now, since we regard M as a submanifold of $\mathbb{R}^d$, g induces a Riemannian metric on $X_n(M)$.

The second metric is defined as follows. Let $x = (x_1, \ldots, x_n) \in X_n(\mathbb{R}^d)$. Let $d(x)$ denote the minimal distance between the points in x, that is:

$$d(x) = \min\{|x_i - x_j| \ : \ i \neq j\},$$

where $|x - y|$ denotes the standard Euclidean distance between vectors $x, y \in \mathbb{R}^d$.

Note that on $X_n(\mathbb{R}^d)$ we have the Euclidean metric restricted from $(\mathbb{R}^d)^n$. We rescale this metric by the factor $\frac{1}{d}$, i.e. we define a metric g_0 on $X_n(\mathbb{R}^d)$ by

$$|v|_{g_0} = \frac{|v|}{d(x)},$$

where $v \in T_x(X_n(\mathbb{R}^d)) = (\mathbb{R}^d)^n$ and $|v|$ is the Euclidean length of $v \in (\mathbb{R}^d)^n$. One should note that $d(x)$, and consequently g_0, is continuous, but not differentiable. A manifold with such a metric is called a C^0-Riemannian manifold. On a C^0-Riemannian manifold one defines the lengths of paths and a geodesic metric in the same way as in the smooth case.

Again, g_0 restricts to a C^0-Riemannian metric on $X_n(M)$.

Proof of Proposition 1.1. We need to show that

$$|[\lambda]|_{P_n(M)} \leq A_0 \cdot l_g(\lambda) + B_0,$$

where λ be a piecewise C^1 loop in $X_n(M)$ based at q, $[\lambda] \in \pi_1(X_n(M), q) \cong P_n(M)$ and $|[\lambda]|_{P_n(M)}$ is the word norm of $[\lambda]$.

By [52, Theorem 4.4] $\overline{X}_n(M)$ is a manifold with corners and $X_n(M)$ is its interior. Moreover, the coordinate charts on $\overline{X}_n(M)$ are defined in such a way that the embedding of $\overline{X}_n(M)$ into $A_n[\mathbb{R}^d]$ is smooth. Thus one can restrict the metric *euc* to $\overline{X}_n(M)$. In other words, $g = \alpha_n^*(euc)$ extends to the compactification $\overline{X}_n(M)$. Now the proposition follows directly from the Schwarz-Milnor lemma [22, 54, 44]. More precisely, consider the following formulation of this result [15, Proposition 8.19].

Lemma 3. *Let a group Γ act properly and discontinuously by isometries on a proper length space X. If the action is cocompact, then Γ is finitely generated and for any choice of base-point $x_0 \in X$, the map $\Gamma \to X$, $h \mapsto h \cdot x_0$, is a quasi-isometry.*

Denote by g_c the metric on $\overline{X}_n(M)$ obtained from *euc*. We apply Lemma 3 to $X = \widetilde{\overline{X}}_n(M)$, $\Gamma = \pi_1(\overline{X}_n(M), \pi(x_0)) \cong \pi_1(X_n(M), \pi(x_0))$, and the length space structure on X being induced by the lift $\widetilde{g_c} = \pi^* g_c$ to X of g_c on $K = X/\Gamma = \overline{X}_n(M)$ by the natural projection map $\pi : X \to K$. Note that $d_{\widetilde{g_c}}(x_0, h \cdot x_0) \leq l_{g_c}(\lambda)$ where λ is any C^1 loop in $X_n(M)$ based at $\pi(x_0)$ representing the element h. Moreover, since g_c is an extension of g, we have $l_{g_c}(\lambda) = l_g(\lambda)$. □

2.3 Proof of Proposition 1.2

We begin with the proof of the following technical result.

Lemma 4. *There exists $C > 0$ depending only on n, such that $|v|_g \leq C \cdot |v|_{g_0}$ for every $v \in T(X_n(M))$.*

Proof. Since α_n is a product map, in order to get a bound on the norm $|v|_g$, we need to bound the norms of the vectors $D\pi_{ij}(v)$ and $Ds_{ijk}(v)$. Let $x = (x_1, \ldots, x_n) \in X_n(M)$ and let $v = (v_1, \ldots, v_n) \in T_x(X_n(M))$. There exists $\epsilon > 0$ such that $x + tv \in X_n(\mathbb{R}^d)$ for every $t < \epsilon$. Note that even though the path $\{x + tv\}_{t \in [0,\epsilon]}$ is not contained in $X_n(M)$, it still represents the tangent vector $v \in T_x(X_n(M)) < T_x(X_n(\mathbb{R}^d))$. We have

$$\begin{aligned} D\pi_{ij}(v) &= \frac{d}{dt}_{|t=0} \pi_{ij}(x + tv) \\ &= \frac{d}{dt}_{|t=0} \frac{x_i - x_j + t(v_i - v_j)}{|x_i - x_j + t(v_i - v_j)|} \\ &= \frac{v_i - v_j}{|x_i - x_j|} - \frac{(x_i - x_j)\langle v_i - v_j, x_i - x_j \rangle}{|x_i - x_j|^3}. \end{aligned}$$

By definition $D\pi_{ij}(v)$ is a vector in $\mathbb{R}^d$ tangent to a $(d-1)$-dimensional sphere at point $\pi_{ij}(x)$. By $|D\pi_{ij}(v)|$ we denote its Euclidean length.

Applying the inequalities $\langle v_i - v_j, x_i - x_j \rangle \leq |v_i - v_j||x_i - x_j|$ and $|v_i - v_j| \leq |v_i| + |v_j|$, we obtain:

$$\begin{aligned} |D\pi_{ij}(v)| &\leq 2\frac{1}{|x_i - x_j|}(|v_i| + |v_j|) \\ &\leq \frac{2}{d(x)}(|v_i| + |v_j|) \\ &\leq \frac{2}{d(x)}(\sum_i |v_i|). \end{aligned}$$

Similarly, for s_{ijk} we compute (already in the map):

$$\begin{aligned} De^{-s_{ijk}}(v) &= \frac{d}{dt}_{|t=0} e^{-s_{ijk}(x+tv)} \\ &= \frac{d}{dt}_{|t=0} e^{-\frac{|x_i-x_j+t(v_i-v_j)|}{|x_i-x_k+t(v_i-v_k)|}} \\ &= \Big[\frac{\langle v_i-v_j, x_i-x_j\rangle}{|x_i-x_j||x_i-x_k|} - \frac{|x_i-x_j|\langle x_i-x_k, v_i-v_k\rangle}{|x_i-x_k|^3}\Big] e^{-\frac{|x_i-x_j|}{|x_i-x_k|}}. \end{aligned}$$

By definition $|Ds_{ijk}(v)|_{g_{exp}} = |De^{-s_{ijk}}(v)|$. Using similar inequalities as before we get:

$$|Ds_{ijk}(v)|_{g_{exp}} \leq \Big[\frac{|v_i|+|v_j|}{|x_i-x_k|} + \frac{(|v_i|+|v_k|)|x_i-x_j|}{|x_i-x_k|^2}\Big] e^{-\frac{|x_i-x_j|}{|x_i-x_k|}}.$$

Applying the inequality $e^{-x} \leq \frac{1}{x}$ to the last term we obtain:

$$\begin{aligned} |Ds_{ijk}(v)|_{g_{exp}} &\leq \Big[\frac{|v_i|+|v_j|}{|x_i-x_k|} + \frac{(|v_i|+|v_k|)|x_i-x_j|}{|x_i-x_k|^2}\Big] \frac{|x_i-x_k|}{|x_i-x_j|} \\ &= \frac{|v_i|+|v_j|}{|x_i-x_j|} + \frac{|v_i|+|v_k|}{|x_i-x_k|} \\ &\leq \frac{1}{d(x)}(2|v_i|+|v_j|+|v_k|) \\ &\leq \frac{2}{d(x)}(\sum_i |v_i|). \end{aligned}$$

Finally, since $\sum_i |v_i| \leq \sqrt{n}|v|$, we get:

$$|D\pi_{ij}(v)| \leq \frac{\sqrt{n}}{d(x)}|v|,$$

$$|Ds_{ijk}(v)|_{g_{exp}} \leq \frac{2\sqrt{n}}{d(x)}|v|.$$

We assume that M is compact, therefore there exists $A > 0$ such that $d(x) \leq A$ for every $x \in X_n(M)$. Now we can bound the pulled-back metric $|\cdot|_g$:

$$\begin{aligned} |v|_g^2 &= |D\iota(v)|^2 + \sum_{i,j} |D\pi_{ij}(v)|^2 + \sum_{i,j,k} |Ds_{ijk}(v)|^2_{g_{exp}} \\ &\leq \frac{A^2}{d(x)^2}|v|^2 + \sum_{i,j} \frac{n}{d(x)^2}|v|^2 + \sum_{i,j,k} \frac{4n}{d(x)^2}|v|^2 \\ &= C\frac{|v|^2}{d(x)^2} = C\cdot|v|^2_{g_0}, \end{aligned}$$

for $C = A^2 + n\binom{n}{2} + 4n\binom{n}{3}$. □

Remark 2.3.1. In the proof of Lemma 4, the inequality $e^{-x} < \frac{1}{x}$ is used to show the bound on the length of Ds_{ijk}. Indeed, the choice of the function $e^{-x}\colon [0,\infty] \to [0,1]$ to define a smooth structure and the metric on $[0,\infty]$ is not completely arbitrary. For a different identification, e.g. $ln(\frac{x+e}{x+1})\colon [0,\infty] \to [0,1]$, we get a different smooth structure on $[0,\infty]$ (in the sense, that the identity map is not smooth) and a metric which is not equivalent to g_{exp}. Then after pull-back by α_n we get a metric on $X_n(M)$ which is not equivalent to g and for this metric Lemma 4 might not hold.

Proof of Proposition 1.2. We need to show that

$$\int_{X_n(M)\setminus Z\cup(\phi_1)^{-1}(Z)} l_g(\lambda(x,\overline{\phi}))\, d\mu^{\otimes n}(x) \leq A_1 \cdot l_1(\overline{\phi}) + B_1,$$

where $\overline{\phi} = \{\phi_t\}$ is an isotopy in $\mathcal{G}$ such that $\phi_0 = 1$.

A similar inequality was proven in [43] for a metric which is equivalent to g_0. Let us first describe this metric. By d_M denote the geodesic metric on M induced by the restriction of the standard Riemannian metric on $\mathbb{R}^d$. Let $x \in X_n(M)$ and $v \in T_x(X_n(M))$. We define

$$|v|_{g_b} = \frac{|v|}{d_M(x)},$$

where $d_M(x) = \min\{d_M(x_i, x_j) : i \neq j\}$ and $|v|$ is the Euclidean length of v seen as a vector in $(\mathbb{R}^d)^n$. The difference between the function d used to define g_0 and d_M is that in d we measure the distance between points x_i and x_j in the ambient space $\mathbb{R}^d$ and in d_M inside M. Since M is compact, clearly d and d_M are equivalent and consequently g_b and g_0 are equivalent.

It follows from [43, Lemma 5.2] that

$$\int_{X_n(M)\setminus Z\cup(\phi_1)^{-1}(Z)} l_{g_b}(\{\phi_t \cdot x\})\, d\mu^{\otimes n}(x) \leq C' \cdot l_1(\overline{\phi}).$$

Recall that $\{\phi_t \cdot x\}$ is a path in $X_n(M)$ (ϕ_t acts on $x \in X_n(M)$ component-wise). Since g_0 and g_b are equivalent, this inequality holds as well for g_0 (with a possibly different constant).

Let us now focus on the metric g. Since the geodesic metric d_g defined by g extends to the compactification of $X_n(M)$, the diameter of $(X_n(M), d_g)$ is finite. We can choose the system of short paths $\gamma(x)$ such that $l_g(\gamma(x)) \leq D$ for some $D > 0$ and every $x \in X_n(M) \setminus Z$.

Thus for every $x \in X_n(M) \setminus Z \cup (\phi_1)^{-1}(Z)$ we have

$$l_g(\lambda(x,\overline{\phi})) \leq l_g(\gamma(x)) + l_g(\{\phi_t \cdot x\}) + l_g(\gamma(\phi_1(x))) \leq 2D + l_g(\{\phi_t \cdot x\}).$$

Finally, due to Lemma 4, we have $l_g(\{\phi_t \cdot x\}) \leq C \cdot l_{g_0}(\{\phi_t \cdot x\})$ and the proposition follows. □

References

[1] Vladimir I. Arnold, *Sur la géométrie différentielle des groupes de Lie de dimension infinie et ses applications a l'hydrodynamique des fluides parfaits*, Ann. Inst. Fourier **16** (1966), 319–361.

[2] Vladimir I. Arnold and Boris A. Khesin, *Topological methods in hydrodynamics*, Applied Mathematical Sciences, vol. 125, Springer-Verlag, 1998.

[3] Scott Axelrod and I. M. Singer, *Chern-Simons perturbation theory. II*, J. Differential Geom. **39** (1994), no. 1, 173–213.

[4] Michel Benaim and Jean-Marc Gambaudo, *Metric properties of the group of area preserving diffeomorphisms*, Trans. Amer. Math. Soc. **353** (2001), no. 11, 4661–4672.

[5] Joan S. Birman, *Mapping class groups and their relationship to braid groups*, Comm. Pure Appl. Math. **22** (1969), 213–238. MR 0243519 (39 #4840)

[6] Michael Brandenbursky, *On quasi-morphisms from knot and braid invariants*, J. Knot Theory Ramifications **20** (2011), no. 10, 1397–1417.

[7] ———, *Quasi-morphisms and L^p-metrics on groups of volume-preserving diffeomorphisms*, J. Topol. Anal. **4** (2012), no. 2, 255–270.

[8] Michael Brandenbursky, *Bi-invariant metrics and quasi-morphisms on groups of Hamiltonian diffeomorphisms of surfaces*, Internat. J. Math. **26** (2015), no. 9, 1550066, 29 pages. MR 3391653

[9] Michael Brandenbursky and Jarek Kędra, *On the autonomous norm on the group of area-preserving diffeomorphisms of the 2-disc*, Algebr. Geom. Topol. **13** (2013), 795–816.

[10] ———, *Quasi-isometric embeddings into diffeomorphism groups*, Groups, Geometry and Dynamics **7** (2013), no. 3, 523–534.

[11] Michael Brandenbursky, Jarek Kedra, and Egor Shelukhin, *On the autonomous norm on the group of Hamiltonian diffeomorphisms of the torus*, Commun. Contemp. Math. **20** (2018), no. 2, 1750042, 27. MR 3730755

[12] Michael Brandenbursky and Michał Marcinkowski, *Entropy and quasimorphisms*, J. Mod. Dyn. **15** (2019), 143–163. MR 3983639

[13] Michael Brandenbursky and Egor Shelukhin, *On the L^p-geometry of autonomous Hamiltonian diffeomorphisms of surfaces*, Math. Res. Lett. **22** (2015), no. 5, 1275–1294.

[14] ———, *The L^p-diameter of the group of area-preserving diffeomorphisms of S^2*, Geom. Topol. **21** (2017), no. 6, 3785–3810.

[15] Martin R. Bridson and André Haefliger, *Metric spaces of non-positive curvature*, Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], vol. 319, Springer-Verlag, Berlin, 1999.

[16] Eugenio Calabi, *On the group of automorphisms of a symplectic manifold*, Problems in analysis (Lectures at the Sympos. in honor of Salomon Bochner, Princeton Univ., Princeton, N.J., 1969), Princeton Univ. Press, 1970, pp. 1–26.

[17] Danny Calegari, *scl*, MSJ Memoirs, vol. 20, Mathematical Society of Japan, Tokyo, 2009.

[18] John Crisp and Bert Wiest, *Quasi-isometrically embedded subgroups of braid and diffeomorphism groups*, Trans. Amer. Math. Soc. **359** (2007), no. 11, 5485–5503.

[19] D. G. Ebin, G. Misiołek, and S. C. Preston, *Singularities of the exponential map on the volume-preserving diffeomorphism group*, Geom. Funct. Anal. **16** (2006), no. 4, 850–868.

[20] David G. Ebin, *Geodesics on the symplectomorphism group*, Geom. Funct. Anal. **22** (2012), no. 1, 202–212.

[21] David G. Ebin and Jerrold Marsden, *Groups of diffeomorphisms and the motion of an incompressible fluid.*, Ann. of Math. (2) **92** (1970), 102–163.

[22] V. Efremovič, *The proximity geometry of Riemannian manifolds*, Uspehi Mat. Nauk **8** (1953), no. 189.

[23] Yakov Eliashberg and Tudor Ratiu, *The diameter of the symplectomorphism group is infinite*, Invent. Math. **103** (1991), no. 2, 327—-340.

[24] Albert Fathi, *Transformations et homéomorphismes préservant la mesure. Systèmes dynamiques minimaux*, Ph.D. thesis, Orsay, 1980.

[25] William Fulton and Robert MacPherson, *A compactification of configuration spaces*, Ann. of Math. (2) **139** (1994), no. 1, 183–225.

[26] J.-M. Gambaudo and E. E. Pécou, *Dynamical cocycles with values in the Artin braid group*, Ergodic Theory Dynam. Systems **19** (1999), no. 3, 627–641.

[27] Jean-Marc Gambaudo and Étienne Ghys, *Enlacements asymptotiques*, Topology **36** (1997), no. 6, 1355–1379.

[28] ———, *Commutators and diffeomorphisms of surfaces*, Ergodic Theory Dynam. Systems **24** (2004), no. 5, 1591–1617.

[29] Jean-Marc Gambaudo and Maxime Lagrange, *Topological lower bounds on the distance between area preserving diffeomorphisms*, Bol. Soc. Brasil. Mat. (N.S.) **31** (2000), no. 1, 9–27.

[30] Charles H. Goldberg, *An exact sequence of braid groups*, Math. Scand. **33** (1973), 69–82. MR 334182

[31] Peter Haïssinsky, *L'invariant de Calabi pour les homéomorphismes quasiconformes du disque*, C. R. Math. Acad. Sci. Paris **334** (2002), no. 8, 635–638.

[32] Ursula Hamenstadt, *Geometry of the mapping class group II: A biautomatic structure*, Preprint arXiv:0912.0137, 2009.

[33] Vincent Humilière, *The Calabi invariant for some groups of homeomorphisms*, J. Symplectic Geom. **9** (2011), no. 1, 107–117.

[34] Tomohiko Ishida, *Quasi-morphisms on the group of area-preserving diffeomorphisms of the 2-disk via braid groups*, Proc. Amer. Math. Soc. Ser. B **1** (2014), 43–51.

[35] Michael Kapovich, *RAAGs in Ham*, Geom. Funct. Anal. **22** (2012), no. 3, 733–755.

[36] Christian Kassel and Vladimir Turaev, *Braid groups*, Graduate Texts in Mathematics, vol. 247, Springer, New York.

[37] Michael Khanevsky, *Quasimorphisms on surfaces and continuity in the Hofer norm*, Preprint, arXiv:1906.08429, 2019.

[38] Boris Khesin and Robert Wendt, *The geometry of infinite-dimensional groups*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics], vol. 51, Springer-Verlag, Berlin, 2009.

[39] Sang-hyun Kim and Thomas Koberda, *Anti-trees and right-angled Artin subgroups of braid groups*, Geom. Topol. **19** (2015), no. 6, 3289–3306.

[40] Maxim Kontsevich, *Feynman diagrams and low-dimensional topology*, First European Congress of Mathematics, Vol. II (Paris, 1992), Progr. Math., vol. 120, Birkhäuser, Basel, 1994, pp. 97–121.

[41] ———, *Operads and motives in deformation quantization*, vol. 48, 1999, Moshé Flato (1937–1998), pp. 35–72.

[42] Alexander Kupers, *There is no topological Fulton-MacPherson compactification*, Preprint, arXiv:2011.14855, 2020.

[43] Michał Marcinkowski, *A short proof that the L^p-diameter of $Diff_0(S, area)$ is infinite*, Preprint arXiv:2010.03000, 2020.

[44] J. Milnor, *A note on curvature and fundamental group*, J. Differential Geometry **2** (1968), 1–7.

[45] Leonid Polterovich, *Floer homology, dynamics and groups*, Morse theoretic methods in nonlinear analysis and in symplectic topology (Dordrecht), NATO Sci. Ser. II Math. Phys. Chem., vol. 217, Springer, 2006, pp. 417—-438.

[46] Pierre Py, *Quasi-morphismes de Calabi et graphe de Reeb sur le tore.*, C. R. Math. Acad. Sci. Paris **343** (2006), no. 5, 323—-328.

[47] ———, *Quasi-morphismes et invariant de Calabi*, Ann. Sci. E'cole Norm. Sup. (4) **39** (2006), no. 1, 177–195.

[48] Egor Shelukhin, *"Enlacements asymptotiques" revisited*, Ann. Math. Qué. **39** (2015), no. 2, 205–208.

[49] Alexander Shnirelman, *The geometry of the group of diffeomorphisms and the dynamics of an ideal incompressible fluid*, Mat. Sb. (N.S.) **128(170)** (1985), no. 1, 82–109, 144.

[50] ———, *Generalized fluid flows, their approximation and applications*, Geom. Funct. Anal. **4** (1994), no. 5, 586–620.

[51] ———, *Microglobal analysis of the Euler equations*, J. Math. Fluid Mech. **7** (2005), no. suppl. 3, S387–S396.

[52] Dev P. Sinha, *Manifold-theoretic compactifications of configuration spaces*, Selecta Math. (N.S.) **10** (2004), no. 3, 391–428.

[53] Steven Smale, *Diffeomorphisms of the 2-sphere*, Proc. Amer. Math. Soc. (1959), no. 10, 621–626.

[54] A. S. Švarc, *A volume invariant of coverings*, Doklady Akademii Nauk SSSR **105** (1955), 32–34.

Michael Brandenbusrky, Department of Mathematics, Ben Gurion University, Be'er Sheva 84105, Israel
E-mail address: brandens@math.bgu.ac.il

Michał Marcinkowski, Institute of Mathematics, Wrocław University, pl. Grunwaldzki 2/4, 50-384 Wrocław, Poland
E-mail address: marcinkow@math.uni.wroc.pl

Egor Shelukhin, Department of Mathematics and Statistics, University of Montreal, C.P. 6128 Succ. Centre-Ville Montreal, QC, H3C 3J7, Canada
E-mail address: shelukhin@dms.umontreal.ca